

Copper One of the chemical elements, copper is a soft, red metal that is an excellent conductor of electricity and heat.

Cosmic Something that belongs, or relates to, the universe.

Crankshaft A metal shaft in a engine that changes the up-and-down motion of pistons into rotary motion to drive wheels.

Density The amount of matter stored within a known volume of a material.

Differential A set of gears that revolve at different speeds, allowing a vehicle to go around corners smoothly.

Direct current (DC) An electrical current, such as that from a battery, that flows in one direction to generate electricity.

Displace The action of moving something out of its normal position—for example, an object in a glass of water can displace the water, causing a spill.

Domesticate The process of improving farming methods (of animals or plants) or living styles (such as dogs in houses) to help and develop the way humans live.

Efficient A machine or system that reaches a high level of productivity.

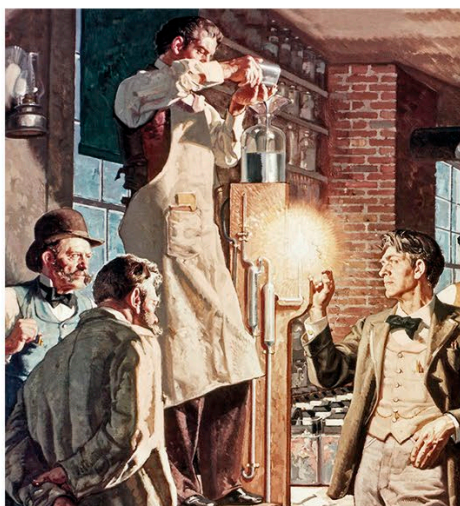
Elasticity A property of a material that allows it to stretch and bend when you push or pull it, and then returns to its original shape.

Electricity A type of energy caused by electrons inside an atom. Static electricity is made by electrons building up in one place, while current electricity happens when electrons move around.

Electrode An electrical contact in a circuit. Electrodes can have a positive or negative charge.

Electromagnet A coil of wire that generates temporary magnetism when electricity flows through it.

Electron This is a particle inside an atom with a negative charge. Electrons orbit the atom's nucleus (or core) in layers called shells.



▲ THOMAS EDISON AND HIS ASSOCIATES TESTING A LAMP

Electronic ignition A system that needs an electronic circuit to start a machine, such as in a car.

Element A pure substance that cannot be broken down into any simpler substances. Elements are the building blocks of all matter. There are 118 elements, most of which occur naturally.

Energy The capacity to do work. It may come from a source such as coal or the sun, and can be used to generate electricity.

Engine A device that burns fuel and oxygen to release stored heat energy that can power a machine.

Equator An imaginary line that goes around the middle of Earth to divide the northern and southern halves of the planet. The equator is often drawn onto maps and globes.

Force A pushing or pulling action that changes an object's speed, direction of movement, or shape.

Fossil fuel A substance that burns easily, releasing heat that has formed from the remains of ancient plants and other organisms. Fossil fuels include coal, natural gas, and oil.

Friction This type of force occurs between two objects, where their surfaces rub against each other to slow down movement.

Fuse A safety device for electrical machines that stops the current flowing if it is too strong.

Gasoline The North American term for petrol—a flammable liquid that is mainly used to fuel different forms of transportation, from cars to boats.

Gearbox The set of gears that connect the engine of a vehicle to its wheels.

Generator A device that converts energy from rotational motion into electricity.

Geocentric A way of viewing or measuring Earth by putting it at the center.

Germ A microscopic plant or animal, especially one that can cause illness.

Gravity The force of attraction that exists between all objects, and is what pulls objects down toward Earth. In space, gravitational forces are much smaller, which is why astronauts float.

Harpoon A barbed, spearlike missile that is used for hunting large fish and whales.

Heliocentric A way of viewing or measuring the sun by putting it at the center.

Hieroglyphics An ancient form of writing that uses pictures to represent sounds and words. The ancient Egyptians, and some other civilizations, used this system.

Hull The main body of a ship or boat, not including the masts, yards, sails, or rigging.

Hydrogen The simplest, lightest, and most abundant of all chemical elements. This gas is one component of water and can be used as a sustainable fuel.

Ignition The system in a gasoline engine that makes the mixture of gasoline and air in the cylinders catch fire.

Incandescent A way of describing something that is very hot and glowing, such as the filament in a light bulb.